Selective Test-Time Reasoning with Relation-Aware Video Evidence for VRR-QA

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Abstract

Video relational reasoning often exhibits a sharp imbalance: many questions can be answered from a compact view of the clip, while a smaller ambiguous subset requires careful inspection across time, viewpoints, and object configurations. We formulate VRR-QA inference as selective verification rather than applying the same computation to every example. The system begins with a direct video-language prediction and measures its stability using several inexpensive alternative views. Consistent cases terminate early; disputed cases receive a dense, timestamped analysis with relation-specific probes and candidate-level checks. Refinement is deliberately conservative: proposing an alternative and authorizing an answer change are handled as separate decisions, and a change is accepted only when temporal visual evidence supports it. The pipeline requires no supervised training or fine-tuning. On the test split, it attains 90.07 average accuracy and 87.81 macro average accuracy, demonstrating that targeted test-time computation can improve relational reasoning without uniformly increasing inference cost.

1. Introduction

A relation visible in one frame may not describe the relation that matters in a video. Apparent left-right position can reverse with viewpoint, depth ordering can be obscured by occlusion, and object motion can be confused with camera motion. Consequently, video relational reasoning requires a model to identify not only entities and actions but also the coordinate frame, temporal extent, and visual evidence that make a relation valid.

VRR-QA was designed around this distinction. Its questions cover spatial, temporal, viewpoint, depth, and visibility relations whose answers may be distributed across several moments rather than stated by a single explicit cue [2]. The benchmark therefore extends the broader shift in video question answering from recognition toward multi-step temporal reasoning [1]. Errors frequently arise when a model commits to a plausible relation before inspecting the frames that disambiguate reference direction, occlusion, or change over time.

Uniformly applying an expensive reasoning procedure is an unattractive solution. A direct video-language model pass is already reliable for many examples, whereas only a limited set exhibits genuine ambiguity. The useful question is thus not simply how to spend more computation, but when additional evidence is likely to change an answer for the right reason. This motivates an inference policy that treats computation as a resource allocated according to prediction stability.

Our pipeline first produces a base answer from a compact video view. Several lightweight views then probe the same question to expose disagreement and generate plausible alternatives. Agreement ends the process immediately. When instability is detected, the example is routed to a dense verifier that records timestamped observations, applies relation-specific visual probes, and directly compares the incumbent answer with its strongest candidate. This hierarchy reserves detailed frame analysis for questions where it can affect the decision.

A central feature of the method is the separation between candidate discovery and answer revision. Disagreement is useful for identifying risky cases, but it is not itself evidence that the base prediction is wrong. The verifier changes an answer only when the alternative is supported by consistent visual and temporal observations. This conservative rule protects reliable first-pass predictions while still recovering difficult cases involving viewpoint, motion, depth, and visibility.

Our contribution is an inference-only framework for adaptive video relational reasoning. It combines stability-based routing, dense relation-aware evidence construction, and evidence-gated answer replacement without supervised training or task-specific fine-tuning. The resulting system is both computationally selective and auditable: extra model calls are concentrated on uncertain examples, and every correction is tied to explicit timestamped evidence.

2. Method

2.1. Overview

Figure 1 shows the full pipeline. The method is an adaptive test-time reasoning system rather than a single prompt. It separates answer production into four nested layers: a direct answer layer, a candidate bank layer, an evidence construc-

tion layer, and a risk-controlled decision layer. The direct layer is applied to all questions. The later layers are activated only when the answer is unstable or when the question asks for a relation that is likely to require explicit temporal evidence.

2.2. Direct Video Inference

The base predictor receives the full video, question, and answer options, and returns a single answer choice. This direct path is the default inference path for all questions. It is simple, reproducible, and avoids unnecessary answer changes when the visual evidence is already clear.

Direct inference is not discarded by later stages. It defines the current answer that every candidate must challenge. This replacement framing is important: a candidate answer is accepted only if it is better supported by visual evidence than the current answer, not merely because it is plausible.

2.3. Candidate Bank Construction

Before verification, the system builds a compact candidate bank. Each question is evaluated under several lightweight views of the same video-question pair. The views differ in frame sampling, storyboard construction, crop or zoom emphasis, temporal alignment, answer-option ordering, and relation-focused prompting. Each view returns an answer and a short evidence statement.

The bank stores the current answer, non-current answer candidates, supporting views, relation keywords, and evidence snippets. It is intentionally recall-oriented: a candidate may enter the bank even if it is not yet safe to apply. Candidate generation and answer replacement are treated as separate problems.

2.4. Disagreement and Hard-Case Routing

The router decides whether a question should remain on the direct path or enter the expensive evidence path. It combines answer instability, candidate support, relation type, and answer-option structure. Questions receive higher priority when independent views disagree, when a non-current answer appears repeatedly, or when the question type is likely to require frame-level evidence:

- lateral, vertical, and front/back spatial relations;
- relative depth and proximity;
- motion direction and trajectory changes;
- viewpoint, visibility, facing direction, and occlusion.

The routing signal is not used as a final answer predictor. It only decides the amount and type of computation assigned to the question. Stable questions keep the direct answer, mild disagreements receive pairwise verification, and hard relation questions receive dense evidence construction.

Table 1. Adaptive evidence schedule for difficult questions. The high-budget path is deliberately more expensive than direct inference, but it is activated only for selected unstable cases.

Stage	Evidence produced	Purpose
Scout	entities, rough timestamps	localize relevant moments
Storyboard views	global frame sequence	preserve temporal context
Crop/zoom views	local entity evidence	resolve small objects and occlusion
Temporal windows	before/after relations	check motion and state changes
Probe matrix	relation-specific facts	test spatial, depth, motion, visibility
Pairwise verifier	current-vs-candidate decision	test replacement safety
Risk gate	ambiguity and contradiction flags	block unsafe corrections

2.5. Adaptive Evidence Scheduler

The high-budget branch uses a fixed sequence of modules, but the number of visual views depends on the routed difficulty. The scheduler starts with a global evidence scout, then expands into dense frame chunks only when the scout finds relevant entities and candidate timestamps. For relation questions, it also requests targeted crops, local zooms, and temporal windows around the suspected decisive moments.

2.6. Dense Frame Evidence Refinement

For a selected difficult question, the video is converted into a sequence of timestamped evidence chunks. Instead of asking the model to immediately choose an answer again, the refinement stage asks for structured observations: which entities are visible, which reference object matters, what relation is visible, which timestamps support each hypothesis, and which ambiguity flags are present.

This produces an explicit evidence table over time. The final decision can therefore be traced to visual observations rather than to a single global model judgment.

2.7. Relation Probe Matrix

The verifier uses a probe matrix specialized to the predicted relation type. Spatial probes check left/right, front/behind, above/below, and reference-frame consistency. Depth probes check foreground/background, occlusion, and proximity. Motion probes compare start and end positions and separate object motion from camera motion. Visibility probes inspect line of sight, facing direction, and blocking objects. Counterfactual probes ask what visual evidence would make the current answer impossible and whether that evidence is present in the clip.

These probes make the reasoning narrower and easier to audit. They also reduce the chance that a visually similar but unsupported candidate wins simply because it matches the broad scene.

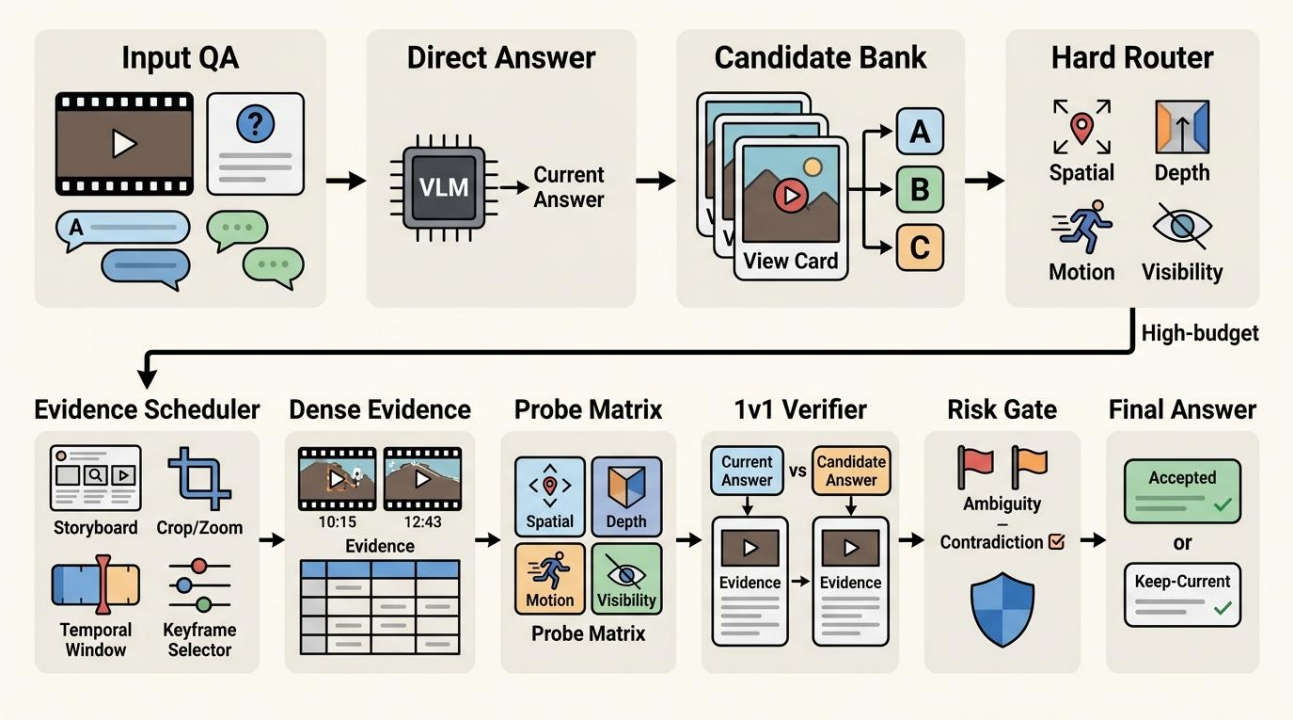


Figure 1. Adaptive Dense Evidence Refinement pipeline. A direct answer is first produced for all questions. Multi-view evidence builds a candidate bank and identifies unstable relation questions. A high-budget branch then schedules dense frame evidence, relation probes, pairwise verification, and risk-gated temporal aggregation before any answer is changed.

2.8. Pairwise and Counterfactual Verification

For each candidate correction, the verifier compares exactly two hypotheses: the current answer and the candidate answer. The prompt asks which hypothesis is better supported by the video evidence and requires evidence for both sides. The verifier is not asked to freely choose any option at this stage.

The pairwise format turns answer refinement into a replacement test. A candidate must identify a visual failure in the current answer and provide timestamped support for itself. If the evidence is weak or ambiguous, the system keeps the current answer.

2.9. Risk-Gated Temporal Aggregation

Frame-level observations are aggregated before any answer is changed. A candidate is accepted only when multiple timestamped observations support it, the current answer is contradicted by the evidence, the support is stable over time or concentrated in a decisive temporal segment, and no high-risk ambiguity flag is present. Otherwise, the direct answer is retained.

The risk gate blocks corrections when the reference object is ambiguous, the relation depends on a single unclear frame, camera motion may explain the apparent trajectory, the relevant entity is partially occluded, or the candidate is supported only by a generic scene match. This conservative rule is critical because raw model disagreement is noisy. The method uses disagreement to focus computation, but it uses dense visual evidence to decide whether a correction is safe.

2.10. Inference-Time Implementation

The system uses no supervised training or fine-tuning. The input is a video clip, a question, and a fixed set of answer options. The direct path produces one current answer for every question. The disagreement detector then evaluates selected lightweight views, including uniformly sampled storyboards, query-guided frames, crop or zoom views, and temporal-alignment prompts.

For hard cases, the dense refinement path builds a timestamped contact sheet and selects relevant frames or chunks. Each chunk is converted into structured fields: visible entities, reference object, observed relation, current-answer support, candidate-answer support, and uncertainty. The final output remains a standard answer file, while the intermediate records form an audit trail for the small set of expensive cases.

Table 2. Test performance of the final system.

Stage	AvgAcc	MacroAvgAcc
Final system	**90.07**	**87.81**

3. Experiments

3.1. Implementation

The system is zero-shot and uses Gemini 3.1 Pro Preview [3] for both direct answering and evidence verification. Direct video calls use temperature 0.0, top-p 0.95, a 2,000-token thinking budget, and 256 output tokens. Evidence calls use 12 uniformly sampled storyboard frames arranged in 4 columns with 320-pixel frame width; visual-review calls use temperature 0.2, top-p 0.95, a 4,000-token thinking budget, and 1,536 output tokens. All answer-producing calls are constrained to the provided options.

3.2. Evaluation Protocol

VRR-QA evaluates multiple-choice video relational reasoning. The system outputs exactly one answer option for each question, and the evaluator reports average accuracy and macro average accuracy. We report the final test configuration only. Additional diagnostic runs are not used as main evidence unless they are executed with the same end-to-end inference protocol as the reported system.

This reporting choice keeps the experiment section aligned with the actual test setting: the direct path is inexpensive and broad, while the dense evidence path is a high-budget route reserved for selected hard cases.

3.3. Test Results

Table 2 reports the test performance of the final system. Scores are percentages. The final system uses direct video inference for stable questions and applies the dense evidence verifier only to routed hard cases.

3.4. Test Ablation

Table 3 compares direct Gemini prompting with evidence-routed systems on the test split. The direct baseline asks the multimodal model to answer from the video context in one pass. The middle configuration verifies candidate answers before replacement, but does not use the full dense evidence schedule. The final system adds dense evidence refinement for routed hard cases. This comparison isolates the value of explicit evidence construction around the same Gemini backbone, while keeping the final output format unchanged.

Table 3. Test ablation of evidence routing with the same Gemini backbone.

Configuration	Inference policy	AvgAcc
Direct Gemini prompting	One video pass	70.47
Simple verification	Candidate check	82.18
Final system	**Dense evidence refinement**	**90.07**

4. Conclusion

We presented Adaptive Dense Evidence Refinement, an inference-only pipeline for VRR-QA. The method uses direct video-language inference for stable questions and reserves high-budget dense evidence analysis for difficult disagreement cases. The key idea is to separate difficult case discovery from answer correction: multiple views identify unstable questions, while dense timestamped evidence and pairwise verification decide whether a candidate should replace the current answer. This risk-controlled design is well suited to video relational reasoning, where many errors depend on spatial reference frames, temporal segments, depth, motion, and visibility.

References

[1] Chaoyou Fu, Yuhan Dai, Yongdong Luo, Lei Li, Shuhuai Ren, Renrui Zhang, Zihan Wang, Chenyu Zhou, Yunhang Shen, Mengdan Zhang, et al. Video-mme: The first-ever comprehensive evaluation benchmark of multi-modal llms in video analysis. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 24108–24118, 2025. 1

[2] Sirnam Swetha, Rohit Gupta, Parth Parag Kulkarni, David G Shatwell, Jeffrey A Chan Santiago, Nyle Siddiqui, Joseph Fioresi, and Mubarak Shah. Vrr-qa: Visual relational reasoning in videos beyond explicit cues. *arXiv preprint arXiv:2506.21742*, 2026. 1

[3] Gemini Team, Rohan Anil, Sebastian Borgeaud, Jean-Baptiste Alayrac, Jiahui Yu, Radu Soricut, Johan Schalkwyk, Andrew M Dai, Anja Hauth, Katie Millican, et al. Gemini: a family of highly capable multimodal models. *arXiv preprint arXiv:2312.11805*, 2023. 4